End-to-End Analysis of Occupational Accidents in Finland

From the StatFin API to Databricks, Power BI and GitHub

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Data Analytics Portfolio Project

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Project Overview

Project objective

The objective was to build a reproducible end-to-end analytics solution to examine occupational accidents in Finland between 2008 and 2024.

Technical objective

- Retrieve data through the StatFin API
- Process the data in Databricks
- Implement a Bronze–Silver–Gold architecture
- Store curated data as Delta tables
- Create an interactive Power BI report
- Document the project on GitHub.

Analytical objective

- Identify long-term trends
- Compare industries
- Examine the severity of the injury.
- Analyze demographic differences
- Communicate the findings clearly

Dataset and Research Questions

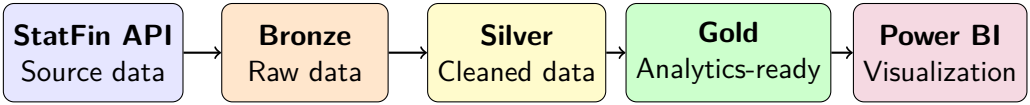
Dataset dimensions

- Year: 2008–2024
- Occupational status
- Industry hierarchy
- Injury category
- Age group
- Gender
- Number of accidents

Research questions

1. How has the number of occupational accidents developed over time?
2. Which industries record the highest accident numbers?
3. How does injury severity vary between industries?
4. What demographic differences can be identified?
5. Which areas may require further investigation?

End-to-End Data Architecture



- **Databricks** handles data extraction, transformation and storage.
- **Delta tables** provide structured and queryable storage between processing layers.
- **Power BI** connects to the Gold layer for analysis and visualization.
- **GitHub** contains the project code, documentation and report screenshots.

Data Extraction from the StatFin API

The StatFin PXWeb API was queried with Python. The request selected the years 2008–2024 and returned the data in JSON-stat2 format.

```
years = [str(year) for year in range(2008, 2025)]

payload = {
    "query": [{
        "code": "Vuosi",
        "selection": {
            "filter": "item",
            "values": years
        }
    }],
    "response": {"format": "json-stat2"}
}

response = requests.post(api_url, json=payload)
response.raise_for_status()
```

Key processing step

The multidimensional JSON-stat2 response was transformed into a tabular structure suitable for Spark processing.

Bronze, Silver and Gold Layers

Bronze

Purpose: Preserve source data

- Raw API response
- Original values retained
- Supports traceability

Silver

Purpose: Clean and standardise

- English column names
- Correct data types
- Cleaned categories

Gold

Purpose: Prepare for analysis

- Industry hierarchy
- Injury categories
- Reporting-ready Delta table

Design principle

Each layer has one clear responsibility: **Bronze preserves**, **Silver cleans**, and **Gold prepares the data for Power BI**.

Important Data Transformations

Transformation challenge

The original industry variable combined industry codes, hierarchy levels, descriptions and special categories in a single field.

Original field

- Industry code
- Industry name
- Mixed hierarchy
- Special values

Transformation

- Extract code
- Identify level
- Separate labels
- Handle exceptions

Gold fields

- `industry_code_main`
- `industry_code_sub`
- `industry_main`
- `industry_sub`

Special cases retained

Special categories, the local government sector and unknown values were handled explicitly instead of being removed.

Power BI Report

Report pages

- Information page
- Overview
- Industry analysis
- Demographic analysis
- Injury severity analysis

Interactive functionality

- Year and industry filters
- KPI cards
- Drill-down through industry levels
- Custom category sorting
- Report-page tooltips

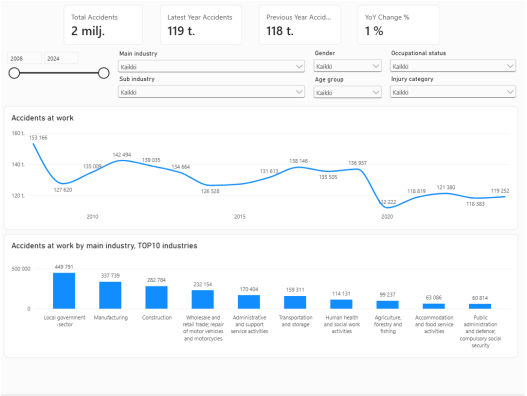


Figure 1: Power BI overview page

Main Findings

1. Long-term development

The annual number of occupational accidents decreased between 2008 and 2024.
The number decreased from approximately 153 000 to 119 000, representing an overall decrease of approximately 22.1%.

2. Largest annual change

The most notable year-on-year change occurred between 2019 and 2020, when accident volume decreased by approximately 18%.

3. Recent development

During the most recent five years, occupational accident volumes have remained relatively stable.

Findings by Industry

Highest accident volume

The highest number of recorded occupational accidents was found in the Local government -sector.
In 2024, this industry represented approximately 24.4% of the accidents analyzed.

Change over time

The most substantial increase was observed in Human health and social work activities (+83.9%), while Manufacturing showed the clearest decline (-59.5%).

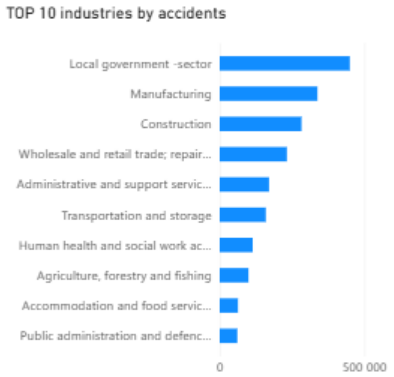


Figure 2: Top 10 industries by accidents

Demographic Findings

Gender

- Men accounted for more recorded accidents.
- The largest gender difference occurred in construction.

Age group

- The highest volumes occurred among 25–34 and 45–54-year-olds.
- The lowest volume occurred among 15–24-year-olds.

Industry differences by age

- Older profile: agriculture, forestry and fishing and the local government sector.
- Younger profile: accommodation and food services and administrative and support services.

These figures describe accident volumes, not individual risk.

Injury Severity Findings

1. Short-duration injuries dominate

In 2024, 66% of occupational accidents resulted in only 0–3 days of absence.

2. Fatal accidents

Fatal accidents were recorded most frequently in transportation and storage and construction.

3. Severity varies with age

Short-duration injuries were the most common across all age groups, but longer-duration injuries represented a larger share in older age groups.

The findings describe the distribution of recorded accidents.

Limitations

1. **Counts are not accident rates**
   The dataset describes the number of accidents but does not directly adjust for the number of employees or hours worked.
2. **Workforce composition affects comparisons**
   Differences by gender, age or industry may partly reflect the size and structure of the underlying workforce.
3. **Classification changes**
   Industry classifications or reporting practices may have changed during the 2008–2024 period.
4. **Aggregated data**
   The data does not support analysis of individual accident circumstances, causes or detailed risk factors.

Skills Demonstrated

Data Engineering

- REST API requests
- JSON-stat processing
- PySpark transformations
- Delta tables
- Medallion architecture

Data Analytics

- Data validation
- Trend analysis
- Category design
- DAX measures
- Contextual interpretation

Communication

- Power BI reporting
- Interactive tooltips
- GitHub documentation
- Visual storytelling
- Limitations and conclusions

Conclusions and Next Steps

Conclusion

The project demonstrates how public statistical data can be transformed into a reproducible analytics solution. It connects data acquisition, engineering, modelling, visualization and communication in a single workflow.

Potential next steps

- Add employment data to calculate accident rates
- Add hours worked as an exposure measure
- Automate annual data refreshes
- Add data-quality tests to the Databricks pipeline
- Compare occupational accidents with economic or labor-market indicators

Questions?

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GitHub: https://github.com/sonjacurry/occupational-accidents-demo
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